

Krishna Khandekar

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Data Scientist –GENAI | Computer Vision | TensorFlow | PyTorch | 3 Years Experience

GenAI Engineer with 3+ years of experience designing and deploying production-grade AI systems across **Computer Vision**, **NLP**, and Retrieval-Augmented Generation (**RAG**). Expertise in CLIP-based semantic search, vector databases (**FAISS**), and scalable ML architectures using PyTorch and Docker. Proven track record of delivering AI-driven automation solutions that reduce manual effort and improve operational efficiency.

EXPERIENCE

Data Scientist

Markgenic Software Pvt. Ltd.

DEC 2022 – Present

- Designed and **deployed RAG-based pipelines** integrating **LLMs, embeddings, and FAISS vector search**, improving retrieval accuracy and response relevance for AI applications.
- Built end-to-end semantic search systems **using CLIP (ViT-B/32) with cosine similarity**, enabling scalable image-to-image retrieval and real-time inference.
- Developed and fine-tuned NLP models (BERT, GRU) for text classification and sentiment analysis, achieving 92% accuracy and improving processing efficiency by 30%.
- Engineered computer vision models (CNN + OpenCV) and deployed ML pipelines using PyTorch, TensorFlow, and Docker, achieving 95%+ accuracy and reducing manual effort by 40%.

SKILLS

- Programming:** Python, SQL
- Machine Learning:** Logistic Regression, Naïve Bayes, XGBoost, Clustering, Predictive Analysis
- Deep Learning:** CNN, RNN, GRU, BERT, GPT
- NLP:** Sentiment Analysis, TF-IDF, Topic Modeling, Text Classification
- GenAI & Retrieval:** Embeddings, FAISS, Vector Search, RAG Architecture
- Computer Vision:** OpenCV, Image Classification, Real-time Detection
- Tools & Deployment:** TensorFlow, PyTorch, Keras, Docker, Git, Jupyter, Django, APIs
- Visualization:** Tableau, Power BI, Excel

PROJECTS

Automated Defect Detection in Manufacturing Using Computer Vision

- Objective:** Developed an automated defect detection system for manufacturing quality control, using Convolutional Neural Networks (CNNs) to identify product defects with over 98% accuracy, significantly reducing the need for manual inspection.
- Data Processing:** Processed and augmented a dataset of 100,000+ labeled product images, applying techniques like image normalization, rotation, and flipping to enhance the model's robustness and improve detection accuracy.
- Model Development:** Implemented a **CNN-based architecture** using **TensorFlow and Keras**, optimized through **hyperparameter tuning** and **transfer learning** techniques, achieving a **35% reduction** in false positives and negatives
- Deployment & Results:** Integrated the model into the production line using **OpenCV** and real-time video analysis, **decreasing defect detection time by 50% and enhancing overall manufacturing efficiency by 20%**

VISUAL PRODUCT INTELLIGENCE (GenAI + CV + Vector Search)

- Engineered **end-to-end image-to-image semantic search system** using **CLIP (ViT-B/32) and FAISS**
- Generated 512-dimensional normalized embeddings and implemented **cosine similarity** retrieval.
- Built high-performance FAISS IndexFlatIP for real-time **top-k nearest neighbor** search.
- Architected modular system extensible to **Image-RAG with LLM-powered** product explanation and **QA**.

EDUCATION

Sinhgad Institute Of Technology Pune, B.E. in Mechanical Engineering | Maharashtra, India(CGPA: 7.68 Jul 2022)

Tech Finalists: Tectonic Robotic Competition (SIT) & Diesel production Project in Mechanical Department

Courses: AI | ML | Python

Certificate: Python, Data Analytics, Data Science

LANGUAGE

English Professional proficiency

Hindi

Native proficiency